Self-Supervised Dueling Networks for Deep Reinforcement Learning

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Deep Reinforcement Learning

Several issues remain in training Deep Reinforcement Learning models:

1. Sparse reward signals.
2. Use of large replay memory in DQN.
3. Slowly updated target network in DQN.

We aim to solve these issues by adding more reward signals into the training phase by introducing auxiliary tasks.

Self-Supervised Learning

Self-supervised learning has seen more attention from the computer vision community. The key is to define surrogate losses and synthesize the annotations from the data itself without labels, allowing for better feature extraction.
For example, it can leverage on the:

1. Spatial relationships between partitions of an image [Doersch et al., 2015], [Noroozi and Favaro, 2016].

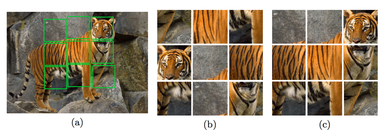

Figure: Using spatial relationships between image regions as a form of self-supervision, from [Noroozi and Favaro, 2016].

2. Temporal relationship between frames of a video [Wang and Gupta, 2015],[Misra et al., 2016].

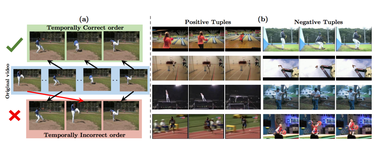

Figure: Using temporal dynamics between video frames as a form of self-supervision, from [Misra et al., 2016].

Since self-supervised methods make use of unannotated data, as auxiliary losses for RL they promise to mine more from the data already available to the policy.

Controllable and Uncontrollable Streams

We propose an alternative approach to auxiliary rewards using a 2 stream architecture:

1. The *controllable* stream that represents state features that our agent can interact with and modify.
2. The *uncontrollable* stream represents the state features from the environment that our agent has no control of.

This separation allows the model to learn action dependent environment interactions and action independent environment dynamics. We also let the model learn to predict the future reward, action and state as extra supervised signals.

Replay Memory Aggregation

In the standard replay memory $\mathcal{D}_1$, each $(s, a, r, s', t) \in \mathcal{D}_1$ represents action dependent state transitions.
In a second replay memory $\mathcal{D}_2$, we sample random states and observe the action independent state transitions. We only have datapoints of the form $(s, a = -1, r, s', t) \in \mathcal{D}_2$, where $a = -1$ indicates no action taken.

Figure: Breakout environment simulation with no action taken.

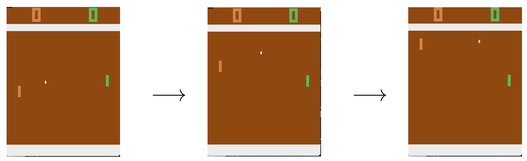

Figure: Pong environment simulation with no action taken.

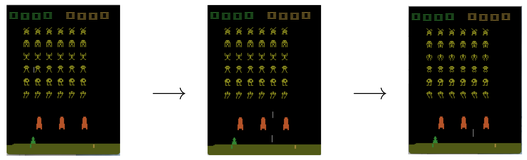

Figure: SpaceInvaders environment simulation with no action taken.

Aggregating the replay memory gives us data to learn features for the *uncontrollable* stream, such as natural movements of the bullets in SpaceInvaders or the ball and opponent in Pong.

Dueling Networks

For our model, since we divide the state representation into two different streams, it is natural for us to follow a similar model design as the Dueling DQN [Wang et al., 2015]. Additional streams on top of the hidden layer are added for self-supervised tasks.

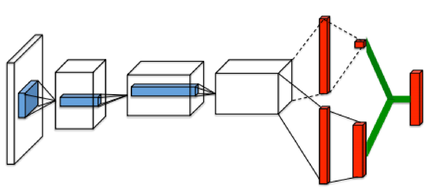

Figure: Dueling Network Architecture.

The q-values are predicted using the Dueling DQN equation:

$$Q(s,a) = V(s) + \Big(A(s,a) - \frac{1}{|\mathcal{A}|}\sum_{a'} A(s,a')\Big)$$

Reward Prediction

The first self-supervised task is reward prediction, a regression problem. We define an auxiliary stream $\mathcal{M}_r$ on top of the last hidden layer of Dueling DQN $\mathcal{M}_{DQN}$. $\mathcal{M}_r$ estimates the reward $\hat{r}$ obtained from being at state s and taking action a. This loss term is as follows:

$$\hat{r}_i = f_r(s, a; \mathcal{M}_r, \mathcal{M}_{DQN})$$
$$\mathcal{L}_r = \frac{1}{N}\sum_i \|r_i - \hat{r}_i\|_2^2$$

for $(s, a, s', r, t) \in \mathcal{D}_1$. We also experimented with predicting a binary classification of whether a reward is received or not by taking action a from state s. The second approach offered slight improvement over the first one.

Best Action prediction

The second self-supervised task is to predict the future best action, a classification problem. We define an auxiliary network $\mathcal{M}_a$ on top of the last hidden layer of Dueling DQN $\mathcal{M}_{DQN}$. $\mathcal{M}_a$ estimates the best action $\hat{a}'$ to choose in the next state s' when the agent is at state s. This loss term is as follows:

$$\hat{a}'_i = f_a(s; \mathcal{M}_a, \mathcal{M}_{DQN})$$
$$\mathcal{L}_a = \frac{1}{N}\sum_i -a'_i \log \hat{a}'_i$$

where $a' = \arg\max_a Q(s', a)$ for $(s, _, s', r, t) \in \mathcal{D}_1$. This method can be viewed as a one step look ahead of what the next best action will be if we let the model run on its own.

Next Frame Regularization

We enforce frames close to each other with no action taken and no reward in between to have similar $Q(s, a)$. Else, we compute compute $Q(s, a)$ using the Bellman equation. This represents the *uncontrollable* stream: the intuition is that if no action is taken and no significant reward is obtained, then two frames close to each other should have similar expected rewards. This loss term is as follows:

$$\mathcal{L}_f = \|Q(s_1, a_1 = -1) - Q(s_2, a_2 = -1)\|_2^2$$

for $(s_1, a_1 = -1, s_2, r = \epsilon, t) \in \mathcal{D}_2$, the aggregated replay memory.

Final Loss Function

Suppose we represent $Q(s, a)$ with a Dueling DQN $\mathcal{M}_{DQN}$ with parameters θ. The final loss function is defined as follows:

$$y_i^{DQN} = r + \gamma \max_{a'} Q(s', a'; \theta^-)$$

$$L_i(\theta_i) = E_{s,a,r,s'}\big[\big(y_i^{DQN} - Q(s, a; \theta_i)\big)^2\big] + \lambda_{\{r,a,f\}}\mathcal{L}_{\{r,a,f\}}$$

where $\lambda_{\{r,a,f\}}$ is a regularization hyperparameter.

Results

We follow the model design and hyperparameters from the original Dueling DQN paper as closely as possible.
The first experiment involves implementing reward prediction and best action prediction on top of the baseline dueling network model. We observe consistent improvement in average reward by including these self-supervised signals.

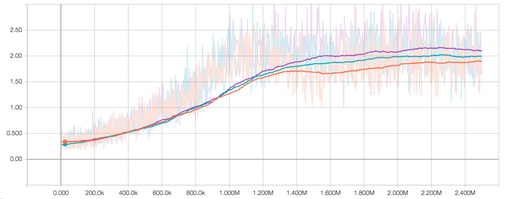

Figure: Results on the Breakout environment. Orange: baseline Dueling DQN. Blue: with reward prediction and best action prediction. Purple: with only reward prediction.

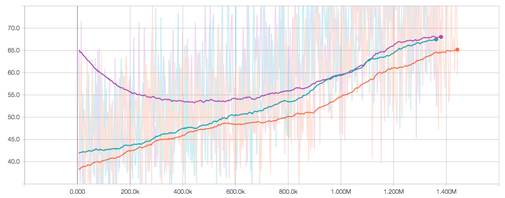

Figure: Results on the SpaceInvaders environment. Orange: baseline Dueling DQN. Blue: with reward prediction and best action prediction. Purple: with only reward prediction.

The second experiment involves implementing next frame regularization on top of the baseline dueling network model. We observe slight improvement in average reward by including this self-supervised signal.

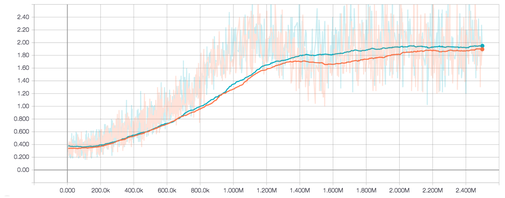

Figure: Results on the SpaceInvaders environment. Orange: baseline Dueling DQN. Blue: with next frame regularization.

Future Work for Report

1. Optimize the next frame regularization to perform better than the baseline Dueling DQN.
2. Visualize the different streams to observe the difference between learning *controllable* and *uncontrollable* streams.
3. Treat the states as a sequence of frames in a video to learn possible temporal dynamics between frames, similar to [Wang and Gupta, 2015].